Using Multilevel QD Projective Method for Solving Multigroup Neutron Transport Coupled with Heat Transfer

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Outline

Project Overview

Problem Definition I

$$\mathcal{L}(T)\psi(\vec{r},\hat{\Omega},E,t) = \mathcal{S}(T)\psi(\vec{r},\hat{\Omega}E,t) + q(\vec{r},\hat{\Omega},E,t) + \sum_k \mathcal{P}_k C_k, \qquad \text{(Neutron Transport)}$$

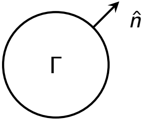
$$\frac{\partial C_k}{\partial t} = -\lambda_k C_k + \int_0^\infty dE\ \beta_k(E)\nu(E)\Sigma_f(E)\int_{4\pi} d\hat{\Omega}\psi(\vec{r},\hat{\Omega},E,t), \qquad \text{(Precursor Balance)}$$

$$\mathcal{H}T(\vec{r}) = \mathcal{Q}(T)\int_{4\pi} d\hat{\Omega}\psi(\vec{r},\hat{\Omega},E,t), \qquad \text{(Heat Transfer)}$$

$$\psi(\vec{r},\Omega,E,0) = \psi^0, \qquad (1d)$$

$$T(\vec{r},0) = T^0, \qquad (1e)$$

$$\psi(\vec{r},\Omega,E,t) = \psi^{in},\ \hat{n}\cdot\hat{\Omega} < 0, \vec{r}\in\partial\Gamma \qquad (1f)$$



Problem Definition II

$$\mathcal{L}(T)\psi = \left[\frac{\partial}{\partial t} + \hat{\Omega}\cdot\nabla + \Sigma_t(T)\right]\psi \tag{2a}$$

$$\mathcal{S}(T)\psi = \frac{\chi(E)}{4\pi}\int_0^\infty dE'\ \nu\Sigma_f(E',T)\int_{4\pi} d\hat{\Omega}\psi(\vec{r},\hat{\Omega},E',t) \tag{2b}$$

$$+\int_0^\infty dE'\int_{4\pi} d\Omega'\ \Sigma_s(\hat{\Omega}'\cdot\hat{\Omega},E'\to E,T)\psi(\hat{\Omega}',E')$$

$$\Sigma_i = \Sigma_{i,0} + \Sigma_{i,1,d}\left(\sqrt{T+\beta_{fb}W_f f(z)} - \sqrt{T_{d,0}}\right) + \Sigma_{i,1,b}\left(T - T_0\right) \tag{2c}$$

$$\mathcal{P}_k C_k = \frac{1}{4\pi}\lambda_k\chi_k^{del}(\vec{r},E,t)C_k(\vec{r},t) \tag{2d}$$

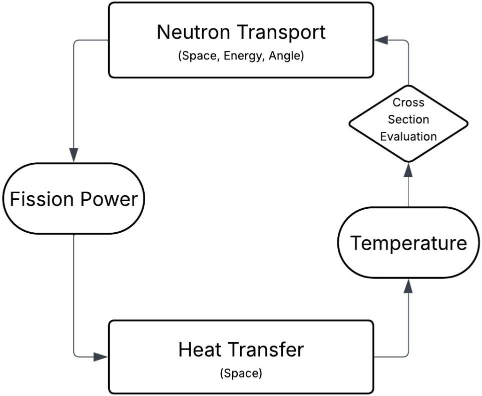
$$\mathcal{Q}(T)\int_{4\pi} d\hat{\Omega}\psi(\vec{r},\hat{\Omega},E,t) = \int_0^\infty dE\ W_f(E)\Sigma_f(E,T)\int_{4\pi} d\hat{\Omega}\psi(\vec{r},\hat{\Omega},E,t) \tag{2e}$$

$$\mathcal{H}T = \left[\frac{\partial}{\partial t} + \dot{m}c_p\nabla - k_b\Delta\right]T(\vec{r}) \tag{2f}$$

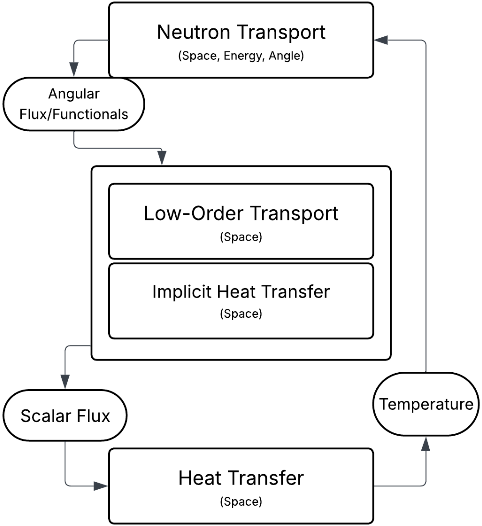
General Coupling Strategy

Coupling HT to High-Order Transport is needlessly expensive

Transport: $\vec{r}, \hat{\Omega}, E, t$
Heat Transfer: $\vec{r}, t$



Coupling with Low-Order Operators



- Implicit HT uses $H\delta T \approx \mathcal{H}\delta T$ to compute cross sections
- Coupling is contained in LO/Approximate system
- Requires few solutions of NT and HT equations
- Introduces no approximation in final solution

CMFD Coupling Methods [1]

CMFD aids iteration by introducing Diffusion Synthetic Acceleration (DSA) and coarse-mesh finite difference (CMFD).

Most apparent coupling method: fixed-point iteration (FPI), which treats cross sections *explicitly*

$$\boxed{\Sigma_i(T^n)} \longrightarrow \boxed{\mathcal{D}\phi^{(n+1)} = \mathcal{S}_0} \longrightarrow \boxed{\mathcal{H}T^{(n+1)} = \mathcal{Q}\phi^{(n+1)}} \qquad \text{(FPI)}$$

$\mathcal{D}$ from transport sweep

CMFD-FPI is an Ineffective method

- Slow to converge
- May be unstable
- Solution of HT equation is expensive

CMFD Coupling Methods – R-CMFD

Instabilities from FPI can be resolved by "relaxing" the heat source on each iteration:

$$\boxed{\Sigma_i(T^n)} \longrightarrow \boxed{\mathcal{D}\phi^{(n+1)} = \mathcal{S}_0} \longrightarrow \boxed{\mathcal{H}T^{(n+1)} = \mathcal{Q}(\alpha\phi^{(n+1)} + (1-\alpha)\phi^{(n)})} \tag{R-CMFD}$$

R-CMFD still has its problems

- Slower to converge than FPI – requires more HT evaluations
- Still may be unstable

CMFD Coupling Methods – X-CMFD

Use implicit cross sections in low-order diffusion calculation

$$\boxed{\mathcal{L}\psi^{(n+1)} = \mathcal{S}\phi^{(n)}} \longrightarrow \boxed{\begin{array}{c} \mathcal{D}(T^{(n+1)})\phi^{(n+1)} = \mathcal{S}_0 \\ \mathcal{H}T^{(n+1)} = \mathcal{Q}\phi^{(n+1)} \end{array}} \tag{X-CMFD}$$

Solved by "inner iterations"

X-CMFD addresses stability

- Stable, converges in same number of outer iterations as single-physics transport
- Requires repeated (expensive) solution of HT

CMFD Coupling Methods – NILO-CMFD

Reduce cost of implicit solution of coupled low-order diffusion and HT

- Use approximation $H \approx \mathcal{H}$ of HT operator to update implicit cross sections within inner iterations
- Update cross sections by Taylor expansion (rather than table lookup or function evaluation)

Implicit cross section update performed during inner iterations:

$$\begin{aligned}
\Sigma^{(n+1)} &= \Sigma(T^{(n+1)}, h^{(n+1)}) && (3)\\
&\approx \Sigma_i^{(n)} + \frac{\partial \Sigma}{\partial T}^{(n)} \left(T^{(n+1)} - T^{(n)} \right) + \frac{\partial \Sigma}{\partial h} \left(h^{(n+1)} - h^{(n)} \right) && (4)\\
&\approx \Sigma_i^{(n)} + \frac{\partial \Sigma}{\partial T}^{(n)} H^{-1} W \left(\Sigma_f \phi^{(n+1)} - \Sigma_f \phi^{(n)} \right) && (5)\\
&\qquad + \frac{\partial \Sigma}{\partial h}^{(n)} W \left(\Sigma_f \phi^{(n+1)} - \Sigma_f \phi^{(n)} \right)
\end{aligned}$$

Multigroup Low-Order Quasidiffusion

Take zeroth and first spatial moments of the high-order multigroup NTE in 1D slab geometry

$$\frac{1}{v_g}\frac{\partial \phi_g}{\partial t} + \frac{\partial J_g}{\partial x} + (\Sigma_{t,g} - \Sigma_{s,g\to g})\phi_g = \sum_{g'=1}^{g-1} \Sigma_{s,g'\to g}\phi_{g'} + \left(Q_{s,g,0} + \chi_g^p \overline{(1-\beta)\nu\Sigma_f}\right)\phi + q_{del,g}^{\chi} \tag{6a}$$

$$\frac{1}{v_g}\frac{\partial J_g}{\partial t} + \frac{\partial}{\partial x}(E_g\phi_g) + \Sigma_{t,g} J_g = 0 \tag{6b}$$

$$\phi_g(x,0) = \phi_g^0, \quad J_g(0,x) = J_g^0 \tag{6c}$$

$$J_g(0,t) = C_{0,g}\phi_g(0,t), \quad J_g(X,t) = C_{X,g}\phi_g(X,t) \tag{6d}$$

Closure terms

QD equations and high-order equations closed by Eddington and boundary factors:

$$E_g = \frac{\int_{-1}^{1} d\mu\, \mu^2 \psi_g}{\int_{-1}^{1} d\mu\, \psi_g}, \tag{7}$$

$$B_{0,g} = \frac{\int_{-1}^{0} d\mu\, \mu \psi_g(0,\mu)}{\int_{-1}^{0} d\mu\, \psi_g(0,\mu)}, \qquad B_{X,g} = \frac{\int_{0}^{1} d\mu\, \mu \psi_g(X,\mu)}{\int_{0}^{1} d\mu\, \psi_g(X,\mu)} \tag{8}$$

Grey Low-Order Quasidiffusion (GLOQD)

Sum the Multigroup LOQD equations to yield

$$\frac{\partial}{\partial t}\left(\frac{\phi}{\overline{v}}\right) + \frac{\partial J}{\partial x} + \left(\overline{\Sigma}_a + \overline{(1-\beta)\nu\Sigma_f}\right)\phi = q_{del}^{\chi} \tag{9a}$$

$$\frac{\partial}{\partial t}\left(\frac{J}{\hat{v}}\right) + \frac{\partial}{\partial x}\left(\overline{E}\phi\right) + \overline{\Sigma}_{tr}J + \overline{\zeta}\phi = 0 \tag{9b}$$

$$\phi(x,0) = \phi^0, \quad J(x,0) = J^0, \tag{9c}$$

$$J(0,t) = \overline{B}_0\phi(0,t), \quad J(X,t) = \overline{B}_X\phi(X,t) \tag{9d}$$

Effect of projection

High-order NTE $\longrightarrow$ MLOQD $\longrightarrow$ GLOQD

$$(x,\mu,E,t) \longrightarrow (x,E,t) \longrightarrow (x,t)$$

Project NTE to same dimensionality as HT

General Project Goals

Combine Advantages of NILO-CMFD [1] and MLOQD [2], write my own implementation

- Couple NTE and HT using Nonlinearly Implicit Low-Order coupling and Multigroup Low-Order Quasidiffusion
- Consider delayed neutrons (not included in [2])
- 1D slab geometry, Linear Discontinuous (LD) finite element method
- HT: advection-diffusion equation using finite volume method

Perform Fourier stability analysis of NILO-MLOQD method

- Ensure stability for all parameters
- Estimate performance of method (iterations saved)
- Compare Fourier results with numerical results

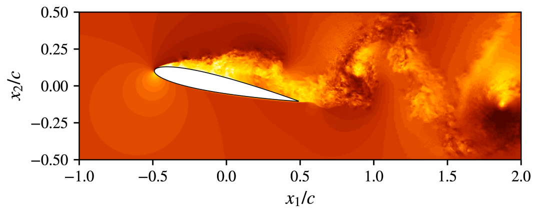
Time, spatial, and angular discretization of ELOT problem

Consider 1D slab/channel geometry with isotropic scattering. Discretize transport using

- Angle – Discrete Ordinates
- Time – Backward Euler
- Space – LD Galerkin Method

and discretize HT using Finite Volume method.

My implementation in Python, but can be scaled up to 2/3D easily using tools such as MFEM [3]



MFEM Navier Minapp (Spieser 2023)

References I

[1] N. J. Adamowicz, A. Manera, and E. W. Larsen, "The NILO-CMFD method for iteratively solving coupled neutron transport–thermal hydraulics problems," *Nuclear Science and Engineering*, vol. 197, no. 2, pp. 262–278, 2023.

[2] A. Tamang and D. Y. Anistratov, "A multilevel projective method for solving the space-time multigroup neutron kinetics equations coupled with the heat transfer equation," *Nuclear Science and Engineering*, vol. 177, no. 1, pp. 1–18, 2014.

[3] R. Anderson, J. Andrej, A. Barker, J. Bramwell, J.-S. Camier, J. Cerveny, V. Dobrev, Y. Dudouit, A. Fisher, T. Kolev, *et al.*, "Mfem: A modular finite element methods library," *Computers & Mathematics with Applications*, vol. 81, pp. 42–74, 2021.